

Course code : **CSE3009**
Course title : **No SQL Data Bases**
Module : **3**
Topic : **5**

Redis Commands

Objectives

This session will give the knowledge about

- Set commands
- SortedSet commands

Set Commands

Sets are unordered collection. Sets do not allow duplicates.

```
> sadd marvel ironman    //sadd to add values into set  
(integer) 1
```

```
> sadd marvel ironman    //set will not allow duplicate  
(integer) 0
```

```
> smembers marvel        //smembers is used to display set elements  
1) "ironman"
```

Set Commands

> smembers marvel

1) "hulk"

2) "thor"

3) "ironman"

> sismember marvel thor
(integer) 1

//sismember is used to check the given
value is a member of set or not

> scard marvel //returns cardinality (size) of the set
3

Set Commands

> smembers marvel

1) "hulk"

2) "thor"

3) "ironman"

> spop marvel //used to remove one or multiple set elements randomly
"thor"

Set Commands

Sunion is to perform union operation between two sets

> smembers marvel

- 1) "thor"
- 2) "hulk"
- 3) "apocalypse"
- 4) "ironman"

> smembers dc

- 1) "aquaman"
- 2) "batman"
- 3) "apocalypse"
- 4) "superman"

> sunion marvel dc

- 1) "aquaman"
- 2) "batman"
- 3) "apocalypse"
- 4) "ironman"
- 5) "thor"
- 6) "hulk"
- 7) "superman"

Set Commands

Sinter is to perform intersection operation between two sets

> smembers marvel

- 1) "thor"
- 2) "hulk"
- 3) "apocalypse"
- 4) "ironman"

> smembers dc

- 1) "aquaman"
- 2) "batman"
- 3) "apocalypse"
- 4) "superman"

> sinter marvel dc

- 1) "apocalypse"

Set Commands

SMOVE source destination member

Moves a member from one set to another

```
> smove marvel dc  
ironman  
(integer) 1
```

```
> smembers marvel  
1) "thor"  
2) "hulk"  
3) "apocalypse"
```

```
> smembers dc  
1) "aquaman"  
2) "batman"  
3) "apocalypse"  
4) "superman"  
5) "ironman"
```


Set Commands

SRANDMEMBER key [count]

Gets one or multiple random members from a set

> smembers marvel

1) "thor"

2) "hulk"

3) "apocalypse"

> srandmember marvel

"thor"

> srandmember marvel

"apocalypse"

Set Commands

SREM key member1 [member2]

Removes one or more specified members from a set

> smembers dc

1) "aquaman"

2) "batman"

3) "apocalypse"

4) "superman"

5) "ironman"

> srem dc ironman

1

Set Commands

If you want to store all the set operation results, add “store” keyword and destination set name

SDIFFSTORE destination key1 [key2]

Sunionstore result marvel dc //perform union and store result into result

Sdiffstore result marvel dc //perform diff and store result into result

Sinterstore result marvel dc //perform inter and store result into result

SortedSet Commands

Redis Sorted Sets are similar to Redis Sets.

The difference is **Sorted Set is associated with a score**, that is used in order to take the sorted set ordered, from the smallest to the greatest score.

ZADD key score1 member1 [score2 member2]

Adds one or more members to a sorted set, or updates its score, if it already exists

```
> zadd courses 1 cse  
(integer) 1
```

SortedSet Commands

> zadd courses 3 mech
(integer) 1

> zadd courses 2 ece
(integer) 1

> zadd courses 2 eee
(integer) 1

ZRANGE key start stop [WITHSCORES]

Returns a range of members in a sorted set,
by index

> zrange courses 0 -1 withscores

- 1) "cse"
- 2) 1.0
- 3) "ece"
- 4) 2.0
- 5) "eee"
- 6) 2.0
- 7) "mech"
- 8) 4.0

SortedSet Commands

ZCARD key

Gets the number of members in a sorted set

```
> zcard courses
```

```
4
```

ZCOUNT key min max

Counts the members in a sorted set with scores within the given values

```
> zcount courses 0 10
```

```
4
```

SortedSet Commands

ZSCORE key member

Gets the score associated with the given member in a sorted set

```
> zscore courses eee  
2.0
```

ZINCRBY key increment member

Increments the score of a member in a sorted set

```
> zincrby courses 2 eee  
4.0
```

SortedSet Commands

ZREVRANGE key start stop
[WITHSCORES]

Returns a range of members in a sorted set, by index, with scores ordered from high to low

```
> zrevrange courses 0 -1
```

1) "eee"

2) "ece"

3) "cse"

ZREVRANGEBYSCORE key max min
[WITHSCORES]

Returns a range of members in a sorted set, by score, with scores ordered from high to low

```
> zrevrangebyscore courses 2 1
```

1) "ece"

2) "cse"

SortedSet Commands

ZREVRANK key member

Determines the index of a member in a sorted set, with scores ordered from high to low

```
> zrevrank courses cse  
2
```

ZSCAN key cursor [MATCH pattern]

[COUNT count]

Incrementally iterates sorted sets elements and associated scores

```
> zscan courses 0
```

1) "0"

2) 1) "cse"

2) 1.0

3) "ece"

4) 2.0

5) "eee"

6) 4.0

SortedSet Commands

ZREM key member [member ...]

Removes one or more members from a sorted set

```
> zrem courses cse
```

```
1
```

```
> zscan courses 0
```

```
1) "0"
```

```
2) 1) "ece"
```

```
2) 2.0
```

```
3) "eee"
```

```
4) 4.0
```

SortedSet Commands

ZREMRANGEBYRANK key start stop

Removes all members in a sorted set within the given indexes

```
> ZADD myzset 1 b 2 c 3 d 4 e
```

```
(integer) 4
```

```
> ZREMRANGEBYRANK myzset 0 2
```

```
(integer) 3
```

```
> ZRANGE myzset 0 -1 WITHSCORES
```

```
1) "e"
```

```
2) "4"
```

SortedSet Commands

ZREMRANGEBYSCORE key min max

Removes all members in a sorted set within the given scores

```
> ZADD myzset 1 b 2 c 3 d 4 e
```

```
(integer) 4
```

```
> ZREMRANGEBYSCORE myzset 2 3
```

```
2
```

```
> ZRANGE myzset 0 -1 WITHSCORES
```

```
1) "b"
```

```
2) 1.0
```

```
3) "e"
```

```
4) 4.0
```

SortedSet Commands

ZINTERSTORE destination numkeys key [key ...]

Intersects multiple sorted sets and stores the resulting sorted set in a new key

> ZADD srcset1 5 M 6 N 7 O

(integer) 3

> ZADD srcset2 3 N 2 O 4 P

(integer) 3

> ZINTERSTORE desset 2 srcset1 srcset2 WEIGHTS 2 3

(integer) 2

> ZRANGE desset 0 -1 WITHSCORES

1) "O"

2) "20"

3) "N"

4) "21"

$$\begin{aligned}
 \text{srcset1} &= \{ (5 \text{ M}), (6 \text{ N}), (7 \text{ O}) \} & \rightarrow \{ (6 * 2, \text{ N}), (7 * 2, \text{ O}) \} \\
 \text{srcset2} &= \{ (3 \text{ N}), (2 \text{ O}), (4 \text{ P}) \} & \rightarrow \{ (3 * 3, \text{ N}), (2 * 3, \text{ O}) \} \\
 \text{desset} &\rightarrow \{ (12 + 9, \text{ N}), (14 + 6, \text{ O}) \} \\
 &= \{ (21, \text{ N}), (20, \text{ O}) \}
 \end{aligned}$$

SortedSet Commands

ZUNIONSTORE destination numkeys key [key ...]

Adds multiple sorted sets and stores the resulting sorted set in a new key

```
> ZADD srcset1 5 M 6 N 7 O
```

```
(integer) 3
```

```
> ZADD srcset2 3 N 2 O 4 P
```

```
(integer) 3
```

```
> ZUNIONSTORE destset 2 srcset1 srcset2 WEIGHTS 2 3
```

```
(integer) 2
```

SortedSet Commands

> ZRANGE desset 0 -1 WITHSCORES

1) "M"

2) "10"

3) "P"

4) "12"

5) "O"

6) "20"

7) "N"

8) "21"

Summary

This session will give the knowledge about

- Set commands
- SortedSet commands
- Reference: <https://redis.io/commands/>
- <https://www.w3resource.com/redis/>